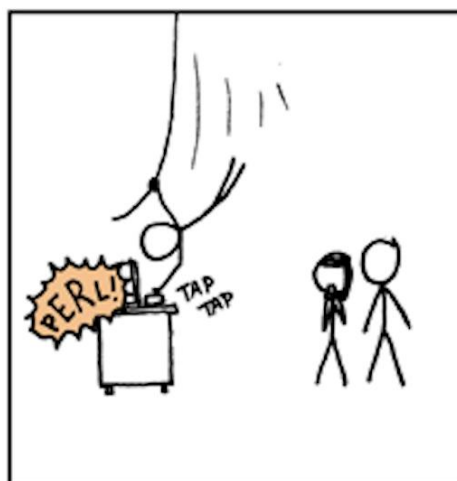
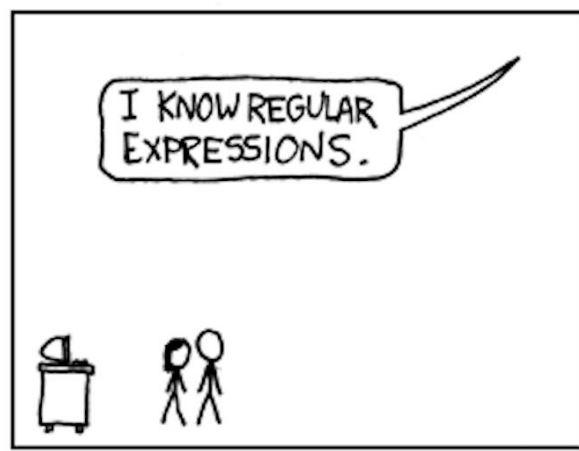
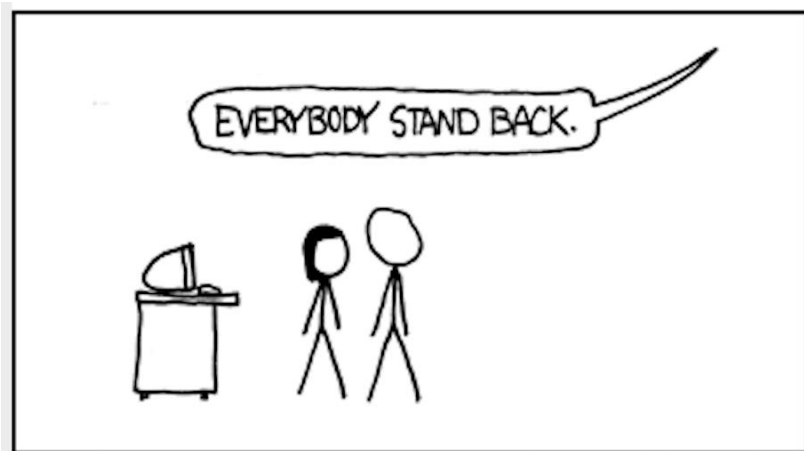


Lexical Analysis

CMPT 379: Compilers

Instructor: Anoop Sarkar

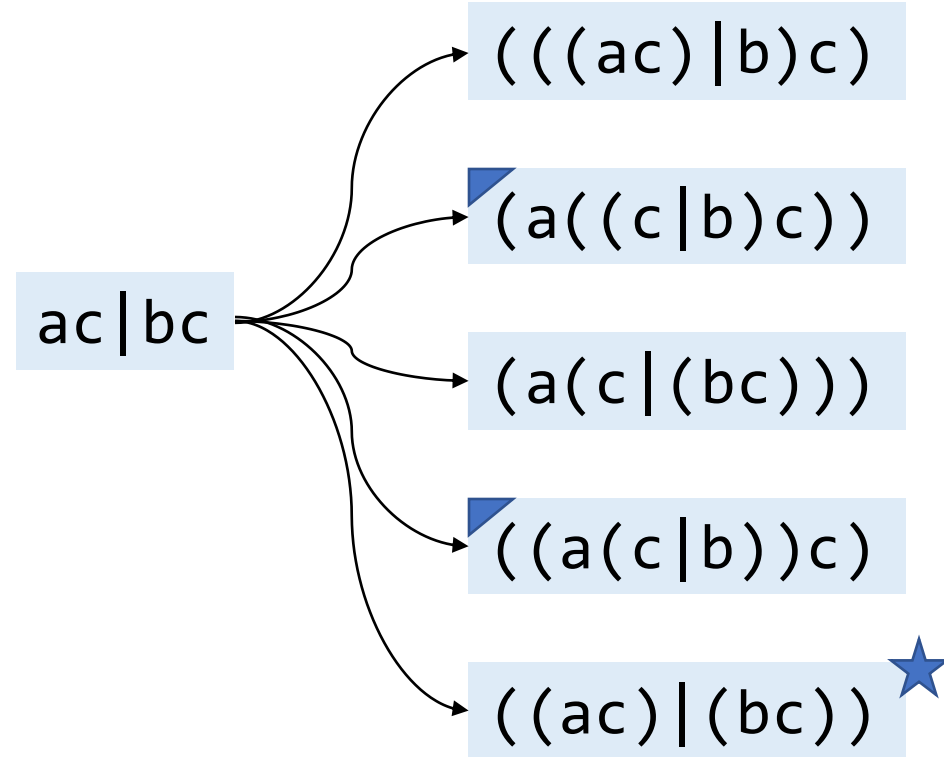
anoopsarkar.github.io/compilers-class



Regular Expressions are Trees

Q: How many possible trees for regexp $ab^*|c^*$

Regular Expressions are ambiguous



Regexp operator precedence rules

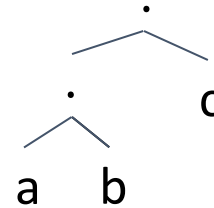
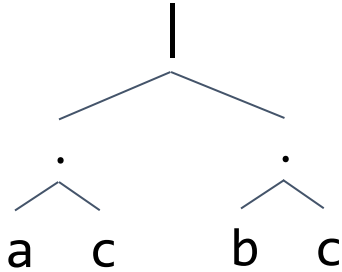
1. Grouping using parentheses ()
2. Unary operator *
3. Binary operator for concatenation
4. Binary operator for alternation |

Q: Find the smallest set of strings that can find the above operator precedence rules for the regexp $ac|bc$.

Hint: Compare the matching on input strings between the original regexp $ac|bc$ and the 5 unambiguous regexps.

Regular expressions are trees

Q: Provide the unambiguous bracketed tree for regexp $ab^*|c^*$ using the precedence and associativity rules



Both alternation and concatenation are commutative, so we use left associativity to get a unique tree.

$ac|bc$

$((ac)|(bc))$

a^{**}

$((a^*)^*)$

abc

$((ab)c)$

Equivalence of Regular Expressions

Equivalence of Regexp

$(0(10)^*1) \mid (01)^* == (01)^* ?$

- $(RS)T == R(ST)$

Commutative

- $(R \mid S) \mid T == R \mid (S \mid T)$

- $(R \mid S) == (S \mid R)$

- $(R \mid S)T == (RT \mid ST)$

Factor

- $R(S \mid T) == RS \mid RT$

- $R == R \mid R$

Redundant

- $R \mid R == R\epsilon$

- $R^* == RR^* \mid \epsilon$

Closure

- $R^*R^* == (R^*)^*$

- $(R^*)^* == R^*$

- $RR^* == R^*R$

Reorder

- $(RS)^*R == R(SR)^*$

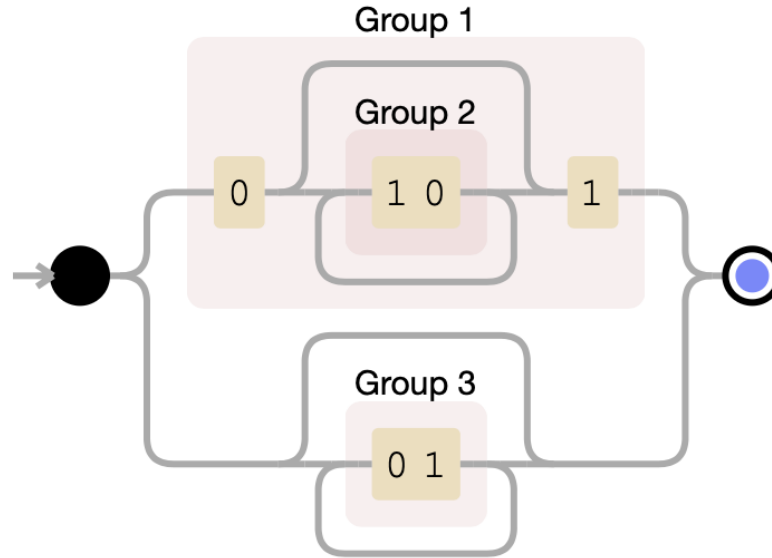
- $(R \mid S)^* == (R^*S^*)^*$

- $(R^*S^*)^* == (R^*S)^*R^*$

- $(R^*S)^*R^* == (R^* \mid S^*)^*$

Equivalence of Regexp

debuggex.com



$(0(10)^*1) \mid (01)^* == (01)^* ?$

Q: Show that $1(0|(01)^*)$ is equivalent to $10|(10)^*1$

Equivalence of Regexp

$$R(A|B) == RA|RB$$

- $0(10)^*1|(01)^*$

$$(RS)^*R == R(SR)^*$$

- $01(01)^*|(01)^*$

- $01(01)^*|(01)^*$

$$RS == (RS)$$

- $01(01)^*|(01)^*$

- $01(01)^*|(01)^*$

$$R+ == RR^*$$

- $01^+|(01)^*$

- $01^+|(01)^*$

$$R+|R^* == (RR^*)|R^* == R^*$$

- 01^*